

# Mathematics Extension 2

## General Instructions

- Reading time – 5 minutes
- Working time – 3 hours
- Write using black pen
- Board-approved calculators may be used
- A table of standard integrals is provided at the back of this paper
- In Questions 11–16, show relevant mathematical reasoning and/or calculations

## Total marks – 100

### Section I Pages 2–6

#### 10 marks

- Attempt Questions 1–10
- Allow about 15 minutes for this section

### Section II Pages 7–18

#### 90 marks

- Attempt Questions 11–16
- Allow about 2 hours and 45 minutes for this section

## Section I

**10 marks**

**Attempt Questions 1–10**

**Allow about 15 minutes for this section**

Use the multiple-choice answer sheet for Questions 1–10.

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**1** Which conic has eccentricity  $\frac{\sqrt{13}}{3}$ ?

(A)  $\frac{x^2}{3} + \frac{y^2}{2} = 1$

(B)  $\frac{x^2}{3^2} + \frac{y^2}{2^2} = 1$

(C)  $\frac{x^2}{3} - \frac{y^2}{2} = 1$

(D)  $\frac{x^2}{3^2} - \frac{y^2}{2^2} = 1$

**2** What value of  $z$  satisfies  $z^2 = 7 - 24i$ ?

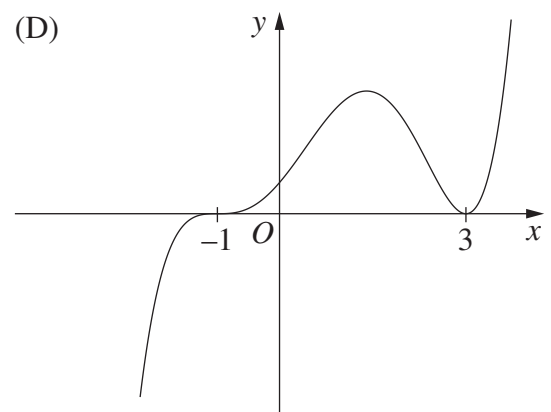
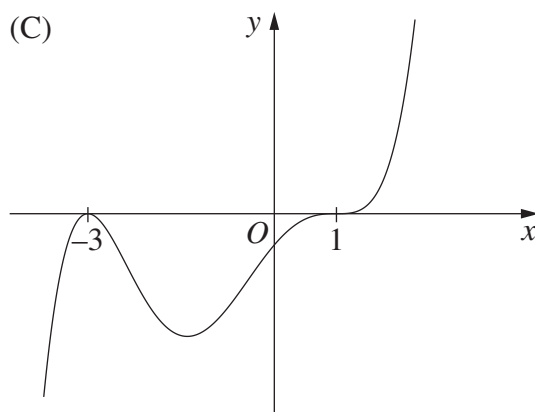
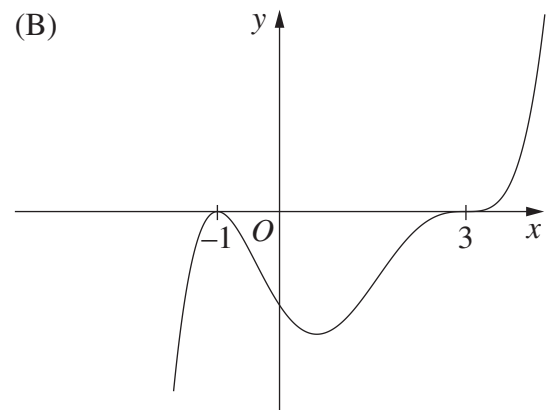
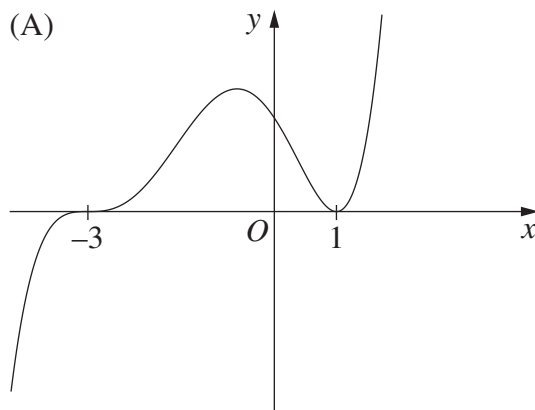
(A)  $4 - 3i$

(B)  $-4 - 3i$

(C)  $3 - 4i$

(D)  $-3 - 4i$

3 Which graph best represents the curve  $y = (x - 1)^2(x + 3)^5$ ?



4 The polynomial  $x^3 + x^2 - 5x + 3$  has a double root at  $x = \alpha$ .

What is the value of  $\alpha$ ?

(A)  $-\frac{5}{3}$

(B)  $-1$

(C)  $1$

(D)  $\frac{5}{3}$

5 Given that  $z = 1 - i$ , which expression is equal to  $z^3$ ?

(A)  $\sqrt{2} \left( \cos\left(\frac{-3\pi}{4}\right) + i \sin\left(\frac{-3\pi}{4}\right) \right)$

(B)  $2\sqrt{2} \left( \cos\left(\frac{-3\pi}{4}\right) + i \sin\left(\frac{-3\pi}{4}\right) \right)$

(C)  $\sqrt{2} \left( \cos\left(\frac{3\pi}{4}\right) + i \sin\left(\frac{3\pi}{4}\right) \right)$

(D)  $2\sqrt{2} \left( \cos\left(\frac{3\pi}{4}\right) + i \sin\left(\frac{3\pi}{4}\right) \right)$

6 Which expression is equal to  $\int x^2 \sin x \, dx$ ?

(A)  $-x^2 \cos x - \int 2x \cos x \, dx$

(B)  $-2x \cos x + \int x^2 \cos x \, dx$

(C)  $-x^2 \cos x + \int 2x \cos x \, dx$

(D)  $-2x \cos x - \int x^2 \cos x \, dx$

7 The numbers  $1, 2, \dots, n$ , for  $n \geq 4$ , are randomly arranged in a row.

What is the probability that the number 1 is somewhere to the left of the number 2?

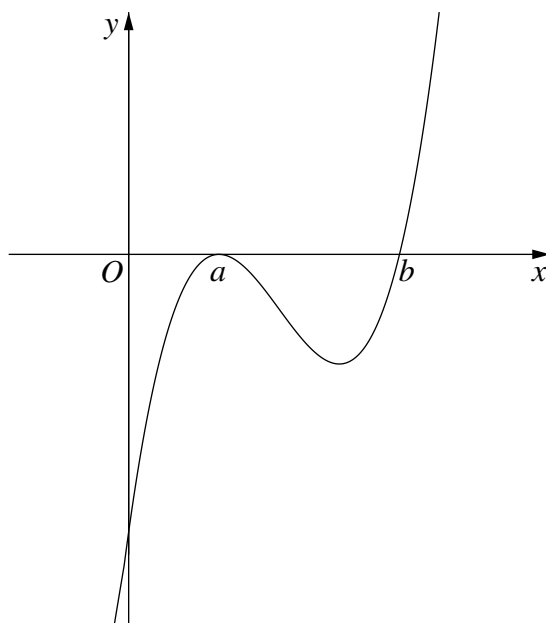
(A)  $\frac{1}{2}$

(B)  $\frac{1}{n}$

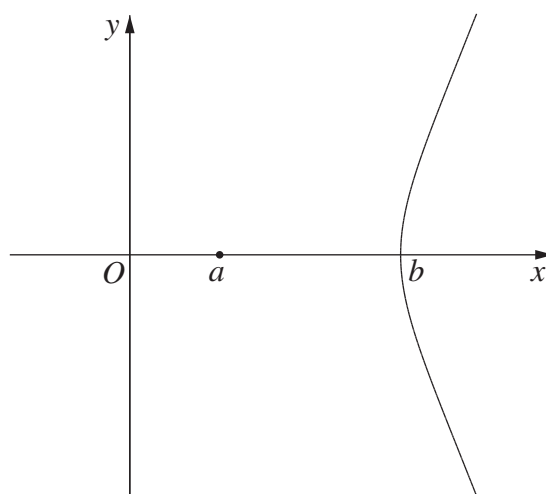
(C)  $\frac{1}{2(n-2)!}$

(D)  $\frac{1}{2(n-1)!}$

- 8 The graph of the function  $y = f(x)$  is shown.



A second graph is obtained from the function  $y = f(x)$ .



Which equation best represents the second graph?

- (A)  $y^2 = |f(x)|$
- (B)  $y^2 = f(x)$
- (C)  $y = \sqrt{f(x)}$
- (D)  $y = f(\sqrt{x})$

- 9 The complex number  $z$  satisfies  $|z - 1| = 1$ .

What is the greatest distance that  $z$  can be from the point  $i$  on the Argand diagram?

- (A) 1
- (B)  $\sqrt{5}$
- (C)  $2\sqrt{2}$
- (D)  $\sqrt{2} + 1$

- 10 Consider the expansion of

$$(1 + x + x^2 + \cdots + x^n)(1 + 2x + 3x^2 + \cdots + (n+1)x^n).$$

What is the coefficient of  $x^n$  when  $n = 100$ ?

- (A) 4950
- (B) 5050
- (C) 5151
- (D) 5253

## Section II

**90 marks**

**Attempt Questions 11–16**

**Allow about 2 hours and 45 minutes for this section**

Answer each question in a SEPARATE writing booklet. Extra writing booklets are available.

In Questions 11–16, your responses should include relevant mathematical reasoning and/or calculations.

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**Question 11** (15 marks) Use a SEPARATE writing booklet.

- (a) Express  $\frac{4+3i}{2-i}$  in the form  $x+iy$ , where  $x$  and  $y$  are real. **2**
- (b) Consider the complex numbers  $z = -\sqrt{3} + i$  and  $w = 3\left(\cos\frac{\pi}{7} + i\sin\frac{\pi}{7}\right)$ .
- (i) Evaluate  $|z|$ . **1**
- (ii) Evaluate  $\arg(z)$ . **1**
- (iii) Find the argument of  $\frac{z}{w}$ . **1**
- (c) Find  $A$ ,  $B$  and  $C$  such that  $\frac{1}{x(x^2+2)} = \frac{A}{x} + \frac{Bx+C}{x^2+2}$ . **2**
- (d) Sketch  $\frac{x^2}{25} + \frac{y^2}{16} = 1$  indicating the coordinates of the foci. **2**
- (e) Find the value of  $\frac{dy}{dx}$  at the point  $(2, -1)$  on the curve  $x + x^2y^3 = -2$ . **3**
- (f) (i) Show that  $\cot\theta + \operatorname{cosec}\theta = \cot\left(\frac{\theta}{2}\right)$ . **2**
- (ii) Hence, or otherwise, find  $\int (\cot\theta + \operatorname{cosec}\theta) d\theta$ . **1**

**Question 12** (15 marks) Use a SEPARATE writing booklet.

- (a) The complex number  $z$  is such that  $|z| = 2$  and  $\arg(z) = \frac{\pi}{4}$ .

Plot each of the following complex numbers on the same half-page Argand diagram.

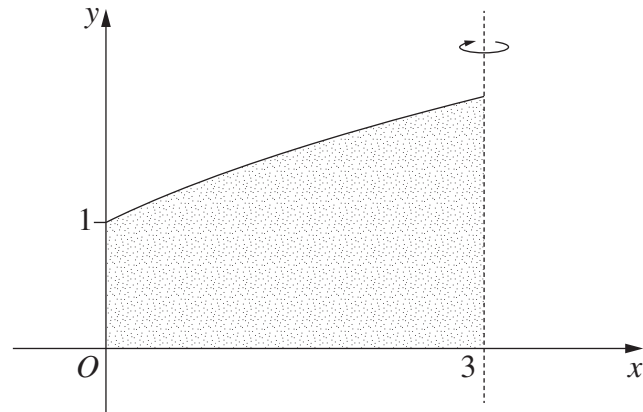
- |       |                     |          |
|-------|---------------------|----------|
| (i)   | $z$                 | <b>1</b> |
| (ii)  | $u = z^2$           | <b>1</b> |
| (iii) | $v = z^2 - \bar{z}$ | <b>1</b> |
- (b) The polynomial  $P(x) = x^4 - 4x^3 + 11x^2 - 14x + 10$  has roots  $a + ib$  and  $a + 2ib$  where  $a$  and  $b$  are real and  $b \neq 0$ .
- |      |                                                                                                        |          |
|------|--------------------------------------------------------------------------------------------------------|----------|
| (i)  | By evaluating $a$ and $b$ , find all the roots of $P(x)$ .                                             | <b>3</b> |
| (ii) | Hence, or otherwise, find one quadratic polynomial with real coefficients that is a factor of $P(x)$ . | <b>1</b> |
- (c) (i) By writing  $\frac{(x-2)(x-5)}{x-1}$  in the form  $mx + b + \frac{a}{x-1}$ , find the equation of the oblique asymptote of  $y = \frac{(x-2)(x-5)}{x-1}$ .
- |      |                                                                                                         |          |
|------|---------------------------------------------------------------------------------------------------------|----------|
|      |                                                                                                         | <b>2</b> |
| (ii) | Hence sketch the graph $y = \frac{(x-2)(x-5)}{x-1}$ , clearly indicating all intercepts and asymptotes. | <b>2</b> |

**Question 12 continues on page 9**

Question 12 (continued)

- (d) The diagram shows the graph  $y = \sqrt{x+1}$  for  $0 \leq x \leq 3$ . The shaded region is rotated about the line  $x = 3$  to form a solid.

4



Use the method of cylindrical shells to find the volume of the solid.

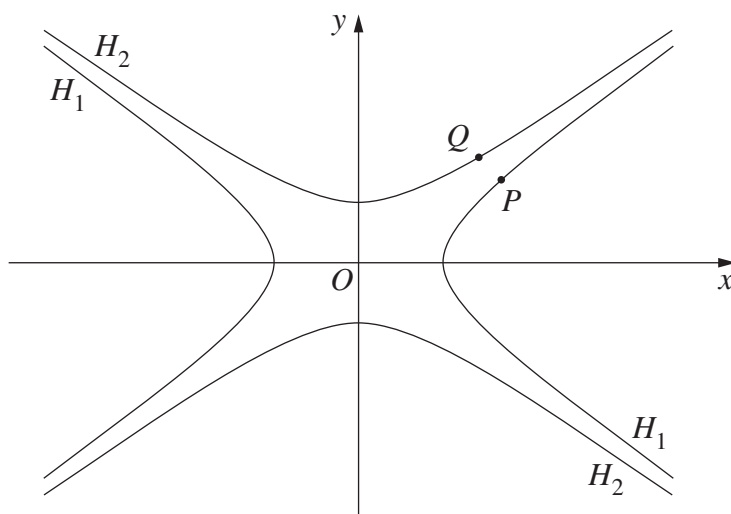
**End of Question 12**

**Question 13** (15 marks) Use a SEPARATE writing booklet.

- (a) The hyperbolas  $H_1 : \frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$  and  $H_2 : \frac{x^2}{a^2} - \frac{y^2}{b^2} = -1$  are shown in the diagram.

Let  $P(a \sec \theta, b \tan \theta)$  lie on  $H_1$  as shown on the diagram.

Let  $Q$  be the point  $(a \tan \theta, b \sec \theta)$ .

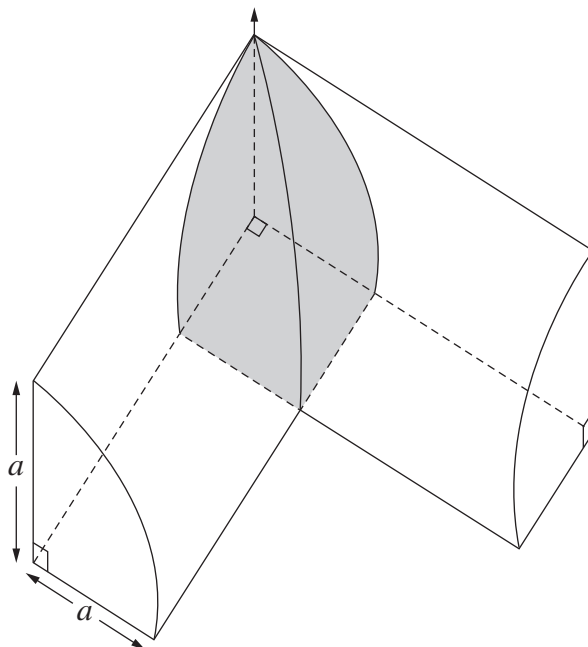


- |                                                                                                       |          |
|-------------------------------------------------------------------------------------------------------|----------|
| (i) Verify that the coordinates of $Q(a \tan \theta, b \sec \theta)$ satisfy the equation for $H_2$ . | <b>1</b> |
| (ii) Show that the equation of the line $PQ$ is $bx + ay = ab(\tan \theta + \sec \theta)$ .           | <b>2</b> |
| (iii) Prove that the area of $\triangle OPQ$ is independent of $\theta$ .                             | <b>3</b> |

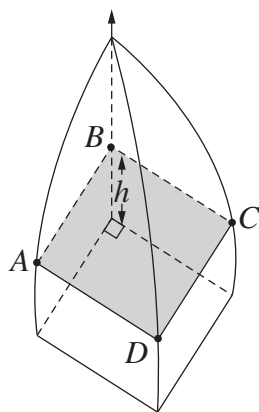
**Question 13 continues on page 11**

Question 13 (continued)

- (b) Two quarter cylinders, each of radius  $a$ , intersect at right angles to form the shaded solid.



A horizontal slice  $ABCD$  of the solid is taken at height  $h$  from the base. You may assume that  $ABCD$  is a square, and is parallel to the base.



- |                                         |          |
|-----------------------------------------|----------|
| (i) Show that $AB = \sqrt{a^2 - h^2}$ . | <b>1</b> |
| (ii) Find the volume of the solid.      | <b>2</b> |

**Question 13 continues on page 12**

Question 13 (continued)

- (c) A small spherical balloon is released and rises into the air. At time  $t$  seconds, it has radius  $r$  cm, surface area  $S = 4\pi r^2$  and volume  $V = \frac{4}{3}\pi r^3$ .

As the balloon rises it expands, causing its surface area to increase at a rate of  $\left(\frac{4\pi}{3}\right)^{\frac{1}{3}} \text{ cm}^2 \text{ s}^{-1}$ . As the balloon expands it maintains a spherical shape.

- (i) By considering the surface area, show that  $\frac{dr}{dt} = \frac{1}{8\pi r} \left(\frac{4}{3}\pi\right)^{\frac{1}{3}}$ . **2**
- (ii) Show that  $\frac{dV}{dt} = \frac{1}{2}V^{\frac{1}{3}}$ . **2**
- (iii) When the balloon is released its volume is  $8000 \text{ cm}^3$ . When the volume of the balloon reaches  $64\,000 \text{ cm}^3$  it will burst. **2**

How long after it is released will the balloon burst?

**End of Question 13**

**Question 14** (15 marks) Use a SEPARATE writing booklet.

(a) (i) Differentiate  $\sin^{n-1}\theta \cos \theta$ , expressing the result in terms of  $\sin \theta$  only. **2**

(ii) Hence, or otherwise, deduce that  $\int_0^{\frac{\pi}{2}} \sin^n \theta d\theta = \frac{(n-1)}{n} \int_0^{\frac{\pi}{2}} \sin^{n-2} \theta d\theta$ , **2**  
for  $n > 1$ .

(iii) Find  $\int_0^{\frac{\pi}{2}} \sin^4 \theta d\theta$ . **1**

(b) The cubic equation  $x^3 - px + q = 0$  has roots  $\alpha$ ,  $\beta$  and  $\gamma$ .

It is given that  $\alpha^2 + \beta^2 + \gamma^2 = 16$  and  $\alpha^3 + \beta^3 + \gamma^3 = -9$ .

(i) Show that  $p = 8$ . **1**

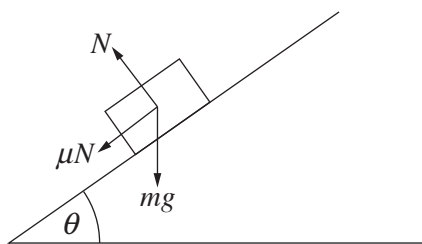
(ii) Find the value of  $q$ . **2**

(iii) Find the value of  $\alpha^4 + \beta^4 + \gamma^4$ . **2**

**Question 14 continues on page 14**

Question 14 (continued)

- (c) A car of mass  $m$  is driven at speed  $v$  around a circular track of radius  $r$ . The track is banked at a constant angle  $\theta$  to the horizontal, where  $0 < \theta < \frac{\pi}{2}$ . At the speed  $v$  there is a tendency for the car to slide up the track. This is opposed by a frictional force  $\mu N$ , where  $N$  is the normal reaction between the car and the track, and  $\mu > 0$ . The acceleration due to gravity is  $g$ .



- (i) Show that  $v^2 = rg \left( \frac{\tan \theta + \mu}{1 - \mu \tan \theta} \right)$ . 3
- (ii) At the particular speed  $V$ , where  $V^2 = rg$ , there is still a tendency for the car to slide up the track. 2

Using the result from part (i), or otherwise, show that  $\mu < 1$ .

**End of Question 14**

**Question 15** (15 marks) Use a SEPARATE writing booklet.

- (a) A particle  $A$  of unit mass travels horizontally through a viscous medium. When  $t = 0$ , the particle is at point  $O$  with initial speed  $u$ . The resistance on particle  $A$  due to the medium is  $kv^2$ , where  $v$  is the velocity of the particle at time  $t$  and  $k$  is a positive constant.

When  $t = 0$ , a second particle  $B$  of equal mass is projected vertically upwards from  $O$  with the same initial speed  $u$  through the same medium. It experiences both a gravitational force and a resistance due to the medium. The resistance on particle  $B$  is  $kw^2$ , where  $w$  is the velocity of the particle  $B$  at time  $t$ . The acceleration due to gravity is  $g$ .

- (i) Show that the velocity  $v$  of particle  $A$  is given by  $\frac{1}{v} = kt + \frac{1}{u}$ . **2**

- (ii) By considering the velocity  $w$  of particle  $B$ , show that **3**

$$t = \frac{1}{\sqrt{gk}} \left( \tan^{-1} \left( u \sqrt{\frac{k}{g}} \right) - \tan^{-1} \left( w \sqrt{\frac{k}{g}} \right) \right).$$

- (iii) Show that the velocity  $V$  of particle  $A$  when particle  $B$  is at rest is given by **1**

$$\frac{1}{V} = \frac{1}{u} + \sqrt{\frac{k}{g}} \tan^{-1} \left( u \sqrt{\frac{k}{g}} \right).$$

- (iv) Hence, if  $u$  is very large, explain why  $V \approx \frac{2}{\pi} \sqrt{\frac{g}{k}}$ . **1**

**Question 15 continues on page 16**

Question 15 (continued)

(b) Suppose that  $x \geq 0$  and  $n$  is a positive integer.

(i) Show that  $1 - x \leq \frac{1}{1+x} \leq 1$ . **2**

(ii) Hence, or otherwise, show that  $1 - \frac{1}{2n} \leq n \ln \left( 1 + \frac{1}{n} \right) \leq 1$ . **2**

(iii) Hence, explain why  $\lim_{n \rightarrow \infty} \left( 1 + \frac{1}{n} \right)^n = e$ . **1**

(c) For positive real numbers  $x$  and  $y$ ,  $\sqrt{xy} \leq \frac{x+y}{2}$ . (Do NOT prove this.)

(i) Prove  $\sqrt{xy} \leq \sqrt{\frac{x^2 + y^2}{2}}$ , for positive real numbers  $x$  and  $y$ . **1**

(ii) Prove  $\sqrt[4]{abcd} \leq \sqrt{\frac{a^2 + b^2 + c^2 + d^2}{4}}$ , for positive real numbers  $a, b, c$  and  $d$ . **2**

**End of Question 15**

**Question 16** (15 marks) Use a SEPARATE writing booklet.

- (a) (i) A table has 3 rows and 5 columns, creating 15 cells as shown. 2

|  |  |  |  |  |
|--|--|--|--|--|
|  |  |  |  |  |
|  |  |  |  |  |
|  |  |  |  |  |

Counters are to be placed randomly on the table so that there is one counter in each cell. There are 5 identical black counters and 10 identical white counters.

Show that the probability that there is exactly one black counter in each column is  $\frac{81}{1001}$ .

- (ii) The table is extended to have  $n$  rows and  $q$  columns. There are  $nq$  counters, where  $q$  are identical black counters and the remainder are identical white counters. The counters are placed randomly on the table with one counter in each cell. 2

Let  $P_n$  be the probability that each column contains exactly one black counter.

Show that  $P_n = \frac{n^q}{\binom{nq}{q}}$ .

- (iii) Find  $\lim_{n \rightarrow \infty} P_n$ . 2

**Question 16 continues on page 18**

Question 16 (continued)

(b) Let  $n$  be a positive integer.

(i) By considering  $(\cos \alpha + i \sin \alpha)^{2n}$ , show that 2

$$\begin{aligned}\cos(2n\alpha) &= \cos^{2n}\alpha - \binom{2n}{2}\cos^{2n-2}\alpha \sin^2\alpha + \binom{2n}{4}\cos^{2n-4}\alpha \sin^4\alpha - \cdots \\ &\quad + \cdots + (-1)^{n-1}\binom{2n}{2n-2}\cos^2\alpha \sin^{2n-2}\alpha + (-1)^n \sin^{2n}\alpha.\end{aligned}$$

Let  $T_{2n}(x) = \cos(2n \cos^{-1} x)$ , for  $-1 \leq x \leq 1$ .

(ii) Show that 2

$$T_{2n}(x) = x^{2n} - \binom{2n}{2}x^{2n-2}(1-x^2) + \binom{2n}{4}x^{2n-4}(1-x^2)^2 + \cdots + (-1)^n(1-x^2)^n.$$

(iii) By considering the roots of  $T_{2n}(x)$ , find the value of 3

$$\cos\left(\frac{\pi}{4n}\right)\cos\left(\frac{3\pi}{4n}\right)\cdots\cos\left(\frac{(4n-1)\pi}{4n}\right).$$

(iv) Prove that 2

$$1 - \binom{2n}{2} + \binom{2n}{4} - \binom{2n}{6} + \cdots + (-1)^n \binom{2n}{2n} = 2^n \cos\left(\frac{n\pi}{2}\right).$$

**End of paper**

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## STANDARD INTEGRALS

$$\int x^n dx = \frac{1}{n+1} x^{n+1}, \quad n \neq -1; \quad x \neq 0, \text{ if } n < 0$$

$$\int \frac{1}{x} dx = \ln x, \quad x > 0$$

$$\int e^{ax} dx = \frac{1}{a} e^{ax}, \quad a \neq 0$$

$$\int \cos ax dx = \frac{1}{a} \sin ax, \quad a \neq 0$$

$$\int \sin ax dx = -\frac{1}{a} \cos ax, \quad a \neq 0$$

$$\int \sec^2 ax dx = \frac{1}{a} \tan ax, \quad a \neq 0$$

$$\int \sec ax \tan ax dx = \frac{1}{a} \sec ax, \quad a \neq 0$$

$$\int \frac{1}{a^2 + x^2} dx = \frac{1}{a} \tan^{-1} \frac{x}{a}, \quad a \neq 0$$

$$\int \frac{1}{\sqrt{a^2 - x^2}} dx = \sin^{-1} \frac{x}{a}, \quad a > 0, \quad -a < x < a$$

$$\int \frac{1}{\sqrt{x^2 - a^2}} dx = \ln \left( x + \sqrt{x^2 - a^2} \right), \quad x > a > 0$$

$$\int \frac{1}{\sqrt{x^2 + a^2}} dx = \ln \left( x + \sqrt{x^2 + a^2} \right)$$

NOTE :  $\ln x = \log_e x, \quad x > 0$

## 2015 HSC Mathematics Extension 2 Marking Guidelines

### Section I

#### Multiple-choice Answer Key

| Question | Answer |
|----------|--------|
| 1        | D      |
| 2        | A      |
| 3        | A      |
| 4        | C      |
| 5        | B      |
| 6        | C      |
| 7        | A      |
| 8        | B      |
| 9        | D      |
| 10       | C      |

## Section II

### Question 11 (a)

| Criteria                                                        | Marks |
|-----------------------------------------------------------------|-------|
| • Provides correct solution                                     | 2     |
| • Attempts to use conjugate of denominator, or equivalent merit | 1     |

*Sample answer:*

$$\begin{aligned}
 \frac{4+3i}{2-i} &= \frac{(4+3i)}{(2-i)} \times \frac{(2+i)}{(2+i)} = \frac{8+6i+4i-3}{4+1} \\
 &= \frac{5+10i}{5} \\
 &= 1+2i
 \end{aligned}$$

### Question 11 (b) (i)

| Criteria                  | Marks |
|---------------------------|-------|
| • Provides correct answer | 1     |

*Sample answer:*

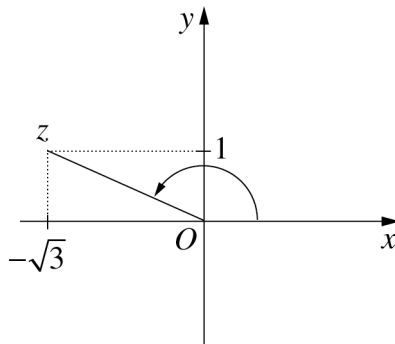
$$\begin{aligned}
 |z| &= |-\sqrt{3}+i| \\
 &= \sqrt{(-\sqrt{3})^2+1} \\
 &= \sqrt{3+1} \\
 &= 2
 \end{aligned}$$

**Question 11 (b) (ii)**

| Criteria                  | Marks |
|---------------------------|-------|
| • Provides correct answer | 1     |

*Sample answer:*

$$\begin{aligned}
 \arg z &= \pi + \tan^{-1}\left(\frac{-1}{\sqrt{3}}\right) \\
 &= \pi - \tan^{-1}\left(\frac{1}{\sqrt{3}}\right) \\
 &= \frac{5\pi}{6}
 \end{aligned}$$

**Question 11 (b) (iii)**

| Criteria                  | Marks |
|---------------------------|-------|
| • Provides correct answer | 1     |

*Sample answer:*

$$\begin{aligned}
 \arg\left(\frac{z}{w}\right) &= \arg z - \arg w \\
 &= \frac{5\pi}{6} - \frac{\pi}{7} \\
 &= \frac{29\pi}{42}
 \end{aligned}$$

**Question 11 (c)**

| Criteria                       | Marks |
|--------------------------------|-------|
| • Provides correct solution    | 2     |
| • Finds A, or equivalent merit | 1     |

*Sample answer:*

$$\frac{1}{x(x^2 + 2)} = \frac{A}{x} + \frac{Bx + C}{x^2 + 2}$$

$$1 = A(x^2 + 2) + (Bx + C)x$$

$$1 = Ax^2 + 2A + Bx^2 + Cx$$

Equating constant terms,

$$1 = 2A$$

$$A = \frac{1}{2}$$

Equating coefficients of  $x^2$ ,

$$0 = A + B$$

$$0 = \frac{1}{2} + B$$

$$B = -\frac{1}{2}$$

Equating coefficients of  $x$ ,

$$C = 0$$

**Question 11 (d)**

| Criteria                                  | Marks |
|-------------------------------------------|-------|
| • Provides correct sketch                 | 2     |
| • Finds eccentricity, or equivalent merit | 1     |

*Sample answer:*For an ellipse  $b^2 = a^2(1 - e^2)$ 

$$\Rightarrow 16 = 25(1 - e^2)$$

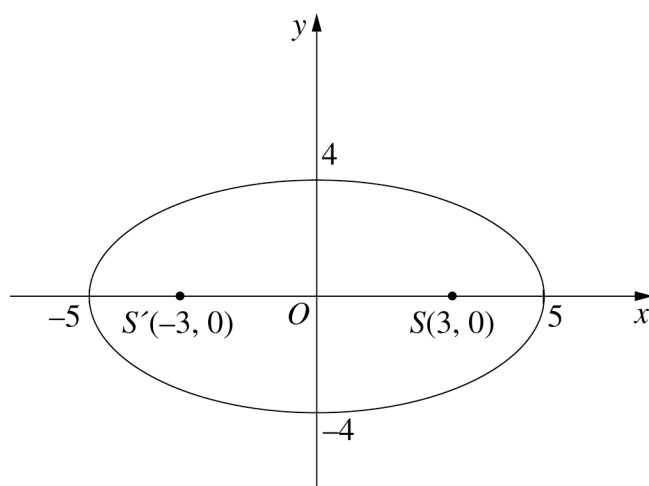
$$\frac{16}{25} = 1 - e^2$$

$$e^2 = 1 - \frac{16}{25}$$

$$e = \frac{3}{5}$$

$$\therefore \text{Foci } (\pm ae, 0) = \left( \pm 5 \times \frac{3}{5}, 0 \right)$$

$$= (\pm 3, 0)$$



**Question 11 (e)**

| Criteria                                        | Marks |
|-------------------------------------------------|-------|
| • Provides correct solution                     | 3     |
| • Correctly differentiates, or equivalent merit | 2     |
| • Uses the chain rule, or equivalent merit      | 1     |

**Sample answer:**Differentiating with respect to  $x$ 

$$1 + 2xy^3 + x^2 3y^2 \frac{dy}{dx} = 0$$

$$\Rightarrow \frac{dy}{dx} = \frac{-1 - 2xy^3}{x^2 3y^2}$$

At  $(2, -1)$ 

$$\begin{aligned} \frac{dy}{dx} &= \frac{-1 - 2 \times 2 \times (-1)^3}{2^2 \times 3 \times (-1)^2} \\ &= \frac{1}{4} \end{aligned}$$

**Question 11 (f) (i)**

| Criteria                                                                                                             | Marks |
|----------------------------------------------------------------------------------------------------------------------|-------|
| • Provides correct solution                                                                                          | 2     |
| • Expresses $\cot \theta$ or $\operatorname{cosec} \theta$ in terms of $\tan \frac{\theta}{2}$ , or equivalent merit | 1     |

**Sample answer:**

$$LHS = \cot \theta + \operatorname{cosec} \theta \quad \text{Let } t = \tan \frac{\theta}{2}$$

$$= \frac{1-t^2}{2t} + \frac{1+t^2}{2t}$$

$$= \frac{2}{2t}$$

$$= \frac{1}{t}$$

$$= \cot \left( \frac{\theta}{2} \right)$$

$$= RHS$$

**Question 11 (f) (ii)**

| Criteria                    | Marks |
|-----------------------------|-------|
| • Provides correct solution | 1     |

*Sample answer:*

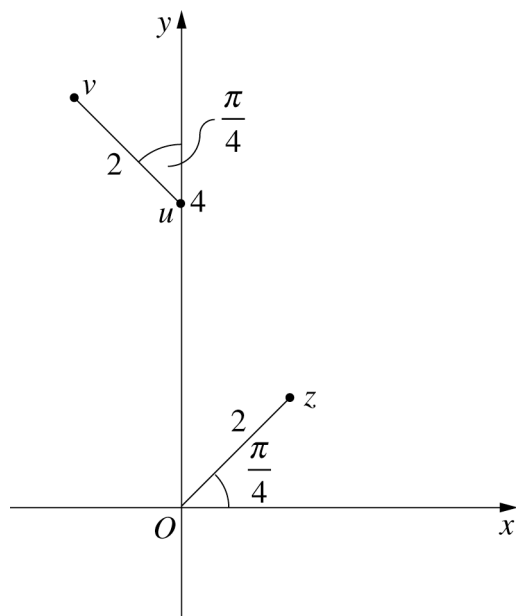
$$\int (\cot \theta + \operatorname{cosec} \theta) d\theta$$

$$= \int \cot \frac{\theta}{2} d\theta$$

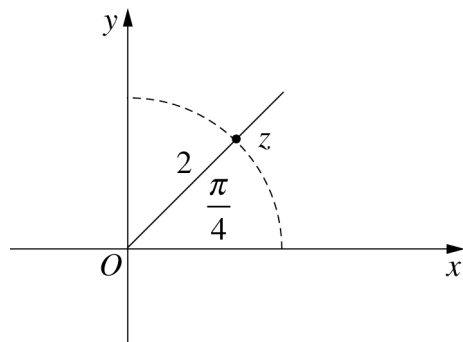
$$= \int \frac{\cos \frac{\theta}{2}}{\sin \frac{\theta}{2}} d\theta$$

$$= \frac{1}{\frac{1}{2}} \ln \left| \sin \frac{\theta}{2} \right| + c$$

$$= 2 \ln \left| \sin \frac{\theta}{2} \right| + c$$

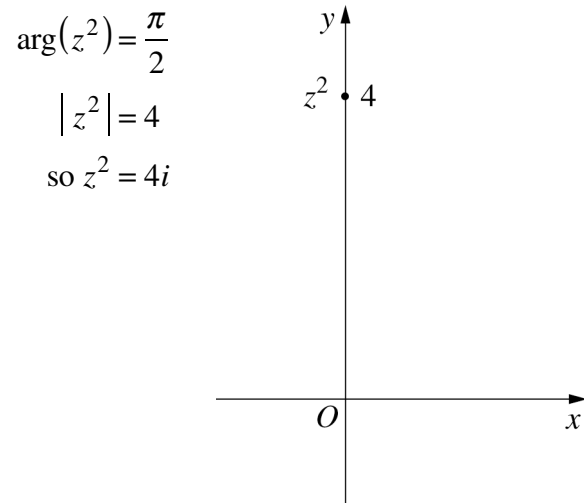
**Question 12 (a) (i), (ii), (iii)****Question 12 (a) (i)**

| Criteria                    | Marks |
|-----------------------------|-------|
| • Provides correct solution | 1     |

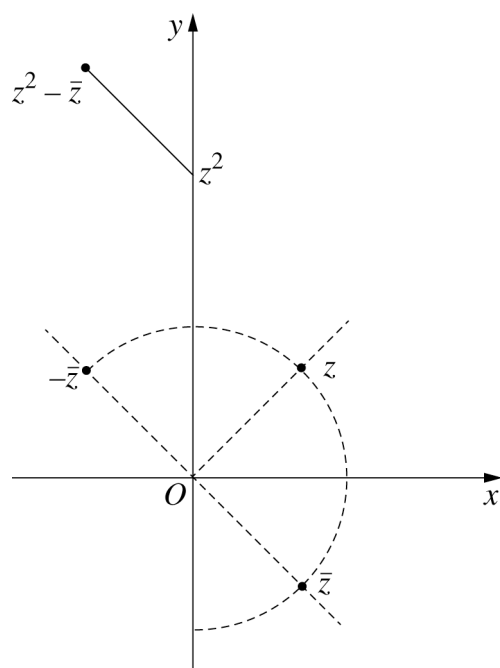
*Sample answer:*

**Question 12 (a) (ii)**

| Criteria                    | Marks |
|-----------------------------|-------|
| • Provides correct solution | 1     |

*Sample answer:***Question 12 (a) (iii)**

| Criteria                    | Marks |
|-----------------------------|-------|
| • Provides correct solution | 1     |

*Sample answer:*

**Question 12 (b) (i)**

| Criteria                                                             | Marks |
|----------------------------------------------------------------------|-------|
| • Provides correct solution                                          | 3     |
| • Finds $a$ , or equivalent merit                                    | 2     |
| • Recognises that the conjugates are also roots, or equivalent merit | 1     |

**Sample answer:**

The roots are  $a \pm ib$  and  $a \pm 2ib$

$\therefore$  The sum of the roots is

$$(a + 2ib) + (a - 2ib) + (a + ib) + (a - ib) = 4$$

$$\Rightarrow 4a = 4$$

$$a = 1$$

The product of the roots is

$$(a + 2ib)(a - 2ib)(a + ib)(a - ib) = 10$$

$$\Rightarrow (a^2 + 4b^2)(a^2 + b^2) = 10$$

But  $a = 1$ ,

$$\therefore (1 + 4b^2)(1 + b^2) = 10$$

$$4b^4 + 5b^2 + 1 - 10 = 0$$

$$4b^4 + 5b^2 - 9 = 0$$

$$(4b^2 + 9)(b^2 - 1) = 0$$

$$b^2 = 1$$

$$b = \pm 1$$

$\therefore$  Roots are  $1 \pm i$ ,  $1 \pm 2i$

**Question 12 (b) (ii)**

| Criteria                    | Marks |
|-----------------------------|-------|
| • Provides correct solution | 1     |

**Sample answer:**

The roots are  $x = 1 \pm i$  and  $x = 1 \pm 2i$

Polynomials with real coefficients have roots that occur in complex conjugate pairs.

$\therefore$  Using the sum and product of the roots, the required quadratics are:

$$\begin{aligned} x^2 - (1 + i + 1 - i)x + (1 + i)(1 - i) & \text{ and } x^2 - (1 + 2i + 1 - 2i)x + (1 + 2i)(1 - 2i) \\ = x^2 - 2x + 2 & \qquad \qquad \qquad = x^2 - 2x + 5 \end{aligned}$$

**Question 12 (c) (i)**

| Criteria                                                  | Marks |
|-----------------------------------------------------------|-------|
| • Provides correct solution                               | 2     |
| • Attempts a division of polynomials, or equivalent merit | 1     |

*Sample answer:*

$$\frac{(x-2)(x-5)}{(x-1)} = \frac{x^2 - 7x + 10}{x-1}$$

Using long division,

$$\begin{array}{r} x-6 \\ x-1 \overline{) x^2 - 7x + 10} \\ \underline{x^2 - x} \phantom{+ 10} \\ -6x + 10 \\ \underline{-6x + 6} \\ 4 \end{array}$$

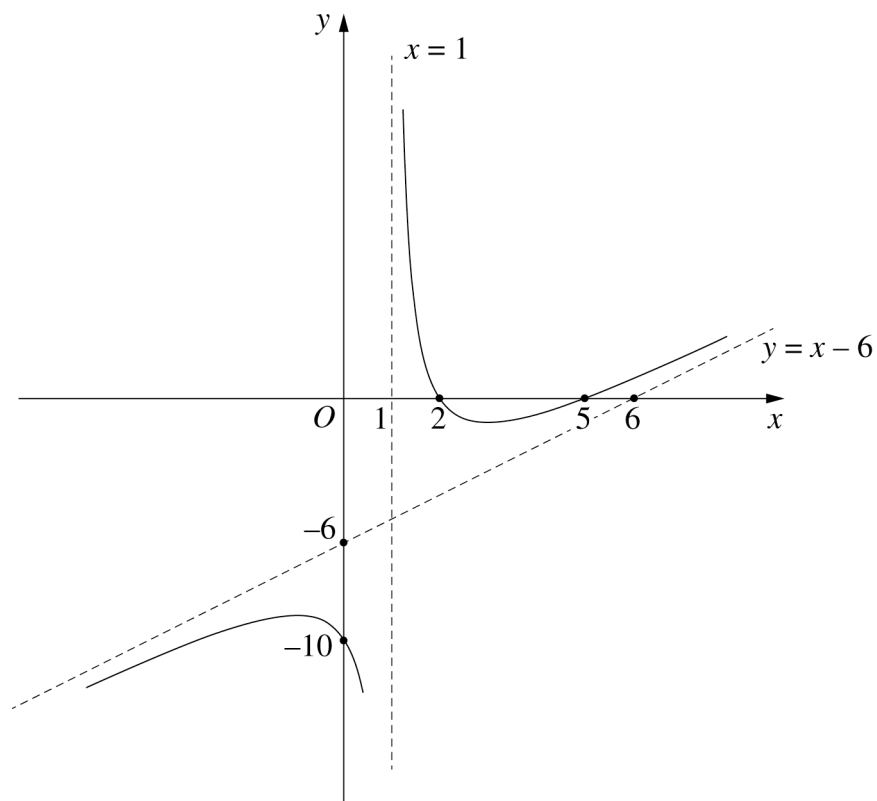
$$\therefore \frac{(x-2)(x-5)}{x-1} = x - 6 + \frac{4}{x-1}$$

where  $m = 1$ ,  $b = -6$ ,  $a = 4$  $\therefore$  The oblique asymptote is  $y = x - 6$

**Question 12 (c) (ii)**

| Criteria                                                                        | Marks |
|---------------------------------------------------------------------------------|-------|
| • Provides correct sketch                                                       | 2     |
| • Identifies any TWO intercepts and the vertical asymptote, or equivalent merit | 1     |

*Sample answer:*



**Question 12 (d)**

| Criteria                                                                                 | Marks |
|------------------------------------------------------------------------------------------|-------|
| • Provides correct solution                                                              | 4     |
| • Attempts to evaluate the correct integral using a suitable method, or equivalent merit | 3     |
| • Provides correct integral expression for the volume, or equivalent merit               | 2     |
| • Recognises the radius is $3 - x$ , or equivalent merit                                 | 1     |

**Sample answer:**

Radius of a typical shell is  $3 - x$

$$\therefore V = 2\pi \int_0^3 (3-x)\sqrt{x+1} dx$$

$$\text{Let } u^2 = x+1$$

$$x = u^2 - 1$$

$$\frac{dx}{du} = 2u$$

$$x = 0, \quad u = 1$$

$$x = 3 \quad u = 2$$

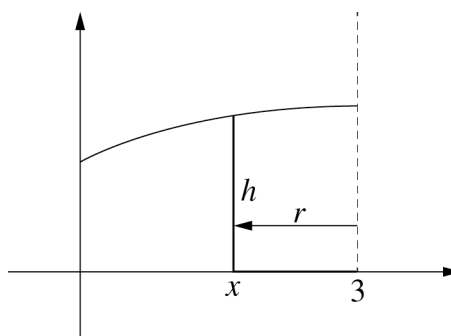
$$\therefore V = 2\pi \int_1^2 (4-u^2)u \cdot 2u du$$

$$= 4\pi \int_1^2 (4u^2 - u^4) du$$

$$= 4\pi \left[ \frac{4u^3}{3} - \frac{u^5}{5} \right]_1^2$$

$$= 4\pi \left( \frac{32}{3} - \frac{32}{5} - \left( \frac{4}{3} - \frac{1}{5} \right) \right)$$

$$= \frac{188\pi}{15} \text{ units}^3$$



area of a typical shell =  $2\pi rh$

**Question 13 (a) (i)**

| Criteria                    | Marks |
|-----------------------------|-------|
| • Provides correct solution | 1     |

**Sample answer:**At  $Q$ ,  $x = a \tan \theta$ ,  $y = b \sec \theta$ , so substituting into  $H_2$ 

$$\begin{aligned}
 LHS &= \frac{(a \tan \theta)^2}{a^2} - \frac{(b \sec \theta)^2}{b^2} \\
 &= \frac{a^2 \tan^2 \theta}{a^2} - \frac{b^2 \sec^2 \theta}{b^2} \\
 &= \frac{a^2 b^2 (\tan^2 \theta - \sec^2 \theta)}{a^2 b^2} \\
 &= \frac{a^2 b^2 (-1)}{a^2 b^2} \quad (\text{since } \tan^2 \theta + 1 = \sec^2 \theta) \\
 &= -1 \\
 &= RHS
 \end{aligned}$$

**Question 13 (a) (ii)**

| Criteria                                        | Marks |
|-------------------------------------------------|-------|
| • Provides correct solution                     | 2     |
| • Finds the slope of $PQ$ , or equivalent merit | 1     |

**Sample answer:**

The gradient of  $PQ$  is  $\frac{b \sec \theta - b \tan \theta}{a \tan \theta - a \sec \theta}$

$$\begin{aligned}
 &= \frac{-b(\tan \theta - \sec \theta)}{a(\tan \theta - \sec \theta)} \\
 &= \frac{-b}{a}
 \end{aligned}$$

 $\therefore$  equation of  $PQ$  is

$$\begin{aligned}
 y - b \tan \theta &= \frac{-b}{a}(x - a \sec \theta) \\
 \Rightarrow ay - ab \tan \theta &= -bx + ab \sec \theta \\
 \Rightarrow bx + ay &= ab(\tan \theta + \sec \theta)
 \end{aligned}$$

**Question 13 (a) (iii)**

| Criteria                                                                    | Marks |
|-----------------------------------------------------------------------------|-------|
| • Provides correct proof                                                    | 3     |
| • Finds a correct expression for the area, or equivalent merit              | 2     |
| • Finds a correct expression for one relevant distance, or equivalent merit | 1     |

**Sample answer:**

$$\begin{aligned}
 PQ^2 &= a^2(\tan \theta - \sec \theta)^2 + b^2(\sec \theta - \tan \theta)^2 \\
 &= (a^2 + b^2)(\tan \theta - \sec \theta)^2 \\
 \therefore PQ &= \sqrt{a^2 + b^2} |\tan \theta - \sec \theta| \\
 &= \sqrt{a^2 + b^2} (\sec \theta - \tan \theta)
 \end{aligned}$$

Since for  $0 \leq \theta \leq \frac{\pi}{2}$ ,  $\sec \theta \geq \tan \theta$

line  $PQ$ :  $bx + ay - ab(\tan \theta + \sec \theta) = 0$

Perpendicular distance from  $O$  to  $PQ$  is

$$\begin{aligned}
 &= \frac{|b \times 0 + a \times 0 - ab(\tan \theta + \sec \theta)|}{\sqrt{a^2 + b^2}} \\
 &= \frac{ab(\tan \theta + \sec \theta)}{\sqrt{a^2 + b^2}}
 \end{aligned}$$

Now, area of  $\triangle OPQ$  is

$$\begin{aligned}
 &\frac{1}{2} \times \sqrt{a^2 + b^2} (\sec \theta - \tan \theta) \times \frac{ab(\tan \theta + \sec \theta)}{\sqrt{a^2 + b^2}} \\
 &= \frac{ab}{2} (\sec^2 \theta - \tan^2 \theta) \\
 &= \frac{ab}{2} (1) \text{ which is independent of } \theta
 \end{aligned}$$

**Question 13 (b) (i)**

| Criteria                    | Marks |
|-----------------------------|-------|
| • Provides correct solution | 1     |

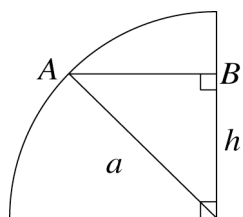
**Sample answer:**

The cylinder is bounded by a circle  $x^2 + y^2 = a^2$

$\therefore$  when  $y = h$ ,  $x = \sqrt{a^2 - h^2}$

Or

Using the right hand quarter cylinder



Using Pythagoras' Theorem,  $AB = \sqrt{a^2 - h^2}$

**Question 13 (b) (ii)**

| Criteria                                                                | Marks |
|-------------------------------------------------------------------------|-------|
| • Provides correct solution                                             | 2     |
| • Finds correct integral expression for the volume, or equivalent merit | 1     |

**Sample answer:**

Area of the slice  $ABCD = a^2 - h^2$

Hence, the volume is given by

$$V = \int_0^a (a^2 - h^2) dh$$

$$= \left[ a^2 h - \frac{h^3}{3} \right]_0^a$$

$$= \frac{2a^3}{3} \text{ units}^3$$

**Question 13 (c) (i)**

| Criteria                                                                            | Marks |
|-------------------------------------------------------------------------------------|-------|
| • Provides correct solution                                                         | 2     |
| • Correctly finds $\frac{dS}{dt}$ in terms of $\frac{dr}{dt}$ , or equivalent merit | 1     |

**Sample answer:**

$$S = 4\pi r^2 \text{ and } \frac{dS}{dt} = \left(\frac{4\pi}{3}\right)^{\frac{1}{3}} \text{ and } \frac{dS}{dt} = \frac{dS}{dr} \cdot \frac{dr}{dt}$$

$$\therefore \frac{dS}{dt} = 8\pi r \frac{dr}{dt}$$

$$\text{so } \left(\frac{4\pi}{3}\right)^{\frac{1}{3}} = 8\pi r \frac{dr}{dt}$$

$$\text{giving } \frac{dr}{dt} = \frac{1}{8\pi r} \left(\frac{4\pi}{3}\right)^{\frac{1}{3}}$$

**Question 13 (c) (ii)**

| Criteria                                                                            | Marks |
|-------------------------------------------------------------------------------------|-------|
| • Provides correct solution                                                         | 2     |
| • Correctly finds $\frac{dV}{dt}$ in terms of $\frac{dr}{dt}$ , or equivalent merit | 1     |

**Sample answer:**

$$\frac{dV}{dt} = \frac{dV}{dr} \cdot \frac{dr}{dt}$$

$$\frac{dV}{dt} = 4\pi r^2 \frac{dr}{dt}$$

$$= 4\pi r^2 \cdot \frac{1}{8\pi r} \left(\frac{4\pi}{3}\right)^{\frac{1}{3}} \text{ by (i)}$$

$$= \frac{1}{2} r \left(\frac{4\pi}{3}\right)^{\frac{1}{3}}$$

$$= \frac{1}{2} \left(r^3 \cdot \frac{4\pi}{3}\right)^{\frac{1}{3}}$$

$$\frac{dV}{dt} = \frac{1}{2} V^{\frac{1}{3}}$$

**Question 13 (c) (iii)**

| Criteria                                                                | Marks |
|-------------------------------------------------------------------------|-------|
| • Provides correct solution                                             | 2     |
| • Finds correct primitive for $t$ in terms of $V$ , or equivalent merit | 1     |

*Sample answer:*

As  $\frac{dV}{dt} = \frac{1}{2}V^{\frac{1}{3}}$

so  $\int \frac{dV}{V^{\frac{1}{3}}} = \frac{1}{2} \int dt$

$$\int_{8\,000}^{64\,000} V^{-\frac{1}{3}} dV = \frac{1}{2} \int_0^t dt$$

$$\therefore \frac{1}{2}t = \left[ \frac{3}{2} V^{\frac{2}{3}} \right]_{8\,000}^{64\,000}$$

$$t = 3(40^2 - 20^2)$$

$$\therefore \text{time} = 3\,600 \text{ seconds}$$

$$(\text{= 1 hour})$$

**Question 14 (a) (i)**

| Criteria                                                                                              | Marks |
|-------------------------------------------------------------------------------------------------------|-------|
| • Provides correct solution                                                                           | 2     |
| • Attempts to differentiate and obtains an expression involving $\cos^2 \theta$ , or equivalent merit | 1     |

**Sample answer:**

$$y = \sin^{n-1} \theta \cos \theta$$

$$y' = u'v + v'u \quad \text{Let } u = \sin^{n-1} \theta \quad v = \cos \theta$$

$$u' = (n-1)\sin^{n-2} \theta \cos \theta \quad v' = -\sin \theta$$

$$y' = (n-1)\sin^{n-2} \theta \cos^2 \theta + \sin^{n-1} \theta (-\sin \theta)$$

$$= (n-1)\sin^{n-2} \theta (1 - \sin^2 \theta) - \sin^n \theta$$

$$= (n-1)\sin^{n-2} \theta - (n-1)\sin^n \theta - \sin^n \theta$$

$$= (n-1)\sin^{n-2} \theta - n\sin^n \theta$$

**Question 14 (a) (ii)**

| Criteria                                                                                 | Marks |
|------------------------------------------------------------------------------------------|-------|
| • Provides correct solution                                                              | 2     |
| • Attempts to evaluate the integral of the expression from part (i), or equivalent merit | 1     |

**Sample answer:**

$$y' = \frac{dy}{d\theta} = (n-1)\sin^{n-2}\theta - n\sin^n\theta$$

$$\int_0^{\frac{\pi}{2}} dy = \int_0^{\frac{\pi}{2}} (n-1)\sin^{n-2}\theta - n\sin^n\theta d\theta$$

$$\left[ y \right]_0^{\frac{\pi}{2}} = (n-1) \int_0^{\frac{\pi}{2}} \sin^{n-2}\theta d\theta - n \int_0^{\frac{\pi}{2}} \sin^n\theta d\theta$$

$$\left[ \sin^{n-1}\theta \cos\theta \right]_0^{\frac{\pi}{2}} = (n-1) \int_0^{\frac{\pi}{2}} \sin^{n-2}\theta d\theta - n \int_0^{\frac{\pi}{2}} \sin^n\theta d\theta$$

$$0 = (n-1) \int_0^{\frac{\pi}{2}} \sin^{n-2}\theta d\theta - n \int_0^{\frac{\pi}{2}} \sin^n\theta d\theta$$

$$n \int_0^{\frac{\pi}{2}} \sin^n\theta d\theta = (n-1) \int_0^{\frac{\pi}{2}} \sin^{n-2}\theta d\theta$$

$$\int_0^{\frac{\pi}{2}} \sin^n\theta d\theta = \left( \frac{n-1}{n} \right) \int_0^{\frac{\pi}{2}} \sin^{n-2}\theta d\theta$$

**Question 14 (a) (iii)**

| Criteria                    | Marks |
|-----------------------------|-------|
| • Provides correct solution | 1     |

*Sample answer:*

$$\begin{aligned}
 \int_0^{\frac{\pi}{2}} \sin^4 \theta d\theta &= \frac{4-1}{4} \int_0^{\frac{\pi}{2}} \sin^2 \theta d\theta \\
 &= \frac{3}{4} \left[ \left( \frac{2-1}{2} \right) \int_0^{\frac{\pi}{2}} d\theta \right] \\
 &= \frac{3}{4} \times \frac{1}{2} \left[ \theta \right]_0^{\frac{\pi}{2}} \\
 &= \frac{3}{8} \left( \frac{\pi}{2} - 0 \right) \\
 &= \frac{3\pi}{16}
 \end{aligned}$$

**Question 14 (b) (i)**

| Criteria                    | Marks |
|-----------------------------|-------|
| • Provides correct solution | 1     |

*Sample answer:*

$$\begin{aligned}
 \alpha + \beta + \gamma &= -\frac{b}{a} = 0 \\
 \alpha\beta + \alpha\gamma + \beta\gamma &= \frac{c}{a} = -p \\
 (\alpha + \beta + \gamma)^2 &= 0 \\
 \text{ie } \alpha^2 + \beta^2 + \gamma^2 + 2(\alpha\beta + \alpha\gamma + \beta\gamma) &= 0 \\
 16 - 2p &= 0 \\
 \therefore p &= 8
 \end{aligned}$$

**Question 14 (b) (ii)**

| Criteria                                                                                                                 | Marks |
|--------------------------------------------------------------------------------------------------------------------------|-------|
| • Provides correct solution                                                                                              | 2     |
| • Obtains a correct equation involving $q$ and at least one of $\alpha^3$ , $\beta^3$ , $\gamma^3$ , or equivalent merit | 1     |

**Sample answer:** $\alpha$  is a root so

$$\alpha^3 = p\alpha - q \quad (1)$$

similarly  $\beta^3 = p\beta - q \quad (2)$

$$\gamma^3 = p\gamma - q \quad (3)$$

adding (1) + (2) + (3)

$$\alpha^3 + \beta^3 + \gamma^3 = p(\alpha + \beta + \gamma) - 3q$$

$$-9 = p \times 0 - 3q$$

$$\therefore q = 3$$

**Question 14 (b) (iii)**

| Criteria                                                                              | Marks |
|---------------------------------------------------------------------------------------|-------|
| • Provides correct solution                                                           | 2     |
| • Obtains an equation involving $\alpha^4 + \beta^4 + \gamma^4$ , or equivalent merit | 1     |

**Sample answer:**

Since  $\alpha^3 = p\alpha - q$

so  $\alpha^4 = p\alpha^2 - q\alpha$

similarly  $\beta^4 = p\beta^2 - q\beta$

$$\gamma^4 = p\gamma^2 - q\gamma$$

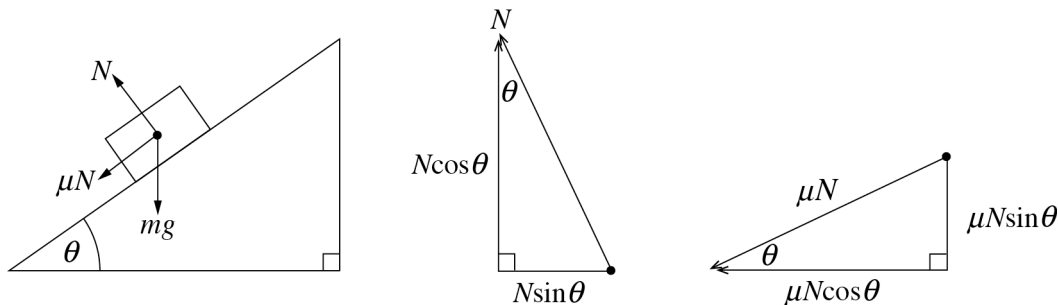
adding  $\alpha^4 + \beta^4 + \gamma^4 = p(\alpha^2 + \beta^2 + \gamma^2) - q(\alpha + \beta + \gamma)$

$$= 8 \times 16 - q \times 0$$

$$= 128$$

**Question 14 (c) (i)**

| Criteria                                                     | Marks |
|--------------------------------------------------------------|-------|
| • Provides correct solution                                  | 3     |
| • Provides correct resolution of forces, or equivalent merit | 2     |
| • Resolves forces in vertical direction, or equivalent merit | 1     |

**Sample answer:**

Resolving forces:

Vertically  $N \cos \theta - \mu N \sin \theta = mg$  (1)

Horizontally  $N \sin \theta + \mu N \cos \theta = \frac{mv^2}{r}$  (2)

Dividing (2) by (1)

$$\frac{\sin \theta + \mu \cos \theta}{\cos \theta - \mu \sin \theta} = \frac{v^2}{rg}$$

$$\therefore v^2 = rg \left( \frac{\sin \theta + \mu \cos \theta}{\cos \theta - \mu \sin \theta} \right)$$

Dividing by  $\frac{\cos \theta}{\cos \theta}$ :

$$v^2 = rg \left( \frac{\tan \theta + \mu}{1 - \mu \tan \theta} \right)$$

**Question 14 (c) (ii)**

| Criteria                                                                               | Marks |
|----------------------------------------------------------------------------------------|-------|
| • Provides correct solution                                                            | 2     |
| • Obtains correct expression for $\tan \theta$ in terms of $\mu$ , or equivalent merit | 1     |

**Sample answer:**Since  $v = V$ 

$$V^2 = rg \text{ and } V^2 = rg \left( \frac{\tan \theta + \mu}{1 - \mu \tan \theta} \right)$$

$$\text{so } \frac{\tan \theta + \mu}{1 - \mu \tan \theta} = 1$$

hence

$$\tan \theta + \mu = 1 - \mu \tan \theta$$

$$\mu + \mu \tan \theta = 1 - \tan \theta$$

$$\text{so } \mu = \frac{1 - \tan \theta}{1 + \tan \theta}$$

$$\text{since } 0 < \theta < \frac{\pi}{2}, \tan \theta > 0$$

$$\text{so } 1 - \tan \theta < 1 \text{ and } 1 + \tan \theta > 1$$

$$\therefore \mu < 1$$

**Question 15 (a) (i)**

| Criteria                                           | Marks |
|----------------------------------------------------|-------|
| • Provides correct solution                        | 2     |
| • Obtains $\ddot{x} = -kv^2$ , or equivalent merit | 1     |

**Sample answer:**

$$0 \quad \xrightarrow{\quad \quad \quad} \quad \quad \quad \longleftarrow -kv^2$$

$$\ddot{x} = -kv^2$$

$$\frac{dv}{dt} = -kv^2$$

$$k dt = -\frac{dv}{v^2}$$

$$\Rightarrow kt = -\int \frac{dv}{v^2}$$

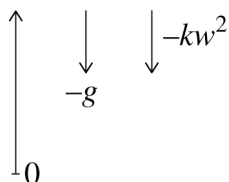
$$kt = \frac{1}{v} + C$$

$$\text{Now } t = 0, v = u \Rightarrow C = -\frac{1}{u}$$

$$\therefore \frac{1}{v} = kt + \frac{1}{u}$$

**Question 15 (a) (ii)**

| Criteria                                                 | Marks |
|----------------------------------------------------------|-------|
| • Provides correct solution                              | 3     |
| • Correctly integrates to find $t$ , or equivalent merit | 2     |
| • Obtains $\ddot{x} = -g - kw^2$ , or equivalent merit   | 1     |

**Sample answer:**

$$\ddot{x} = -g - kw^2$$

$$\therefore \frac{dw}{dt} = -(g + kw^2)$$

$$\frac{dt}{dw} = -\frac{1}{g + kw^2}$$

$$t = -\int \frac{1}{g + kw^2} dw$$

$$= -\frac{1}{k} \int \frac{1}{\frac{g}{k} + w^2} dw$$

$$= -\frac{1}{k} \left[ \sqrt{\frac{k}{g}} \tan^{-1} \left( w \sqrt{\frac{k}{g}} \right) \right] + C$$

$$\therefore t = -\frac{1}{\sqrt{gk}} \tan^{-1} \left( w \sqrt{\frac{k}{g}} \right) + C$$

Now  $t = 0$ ,  $w = u$ , so

$$C = \frac{1}{\sqrt{gk}} \tan^{-1} \left( u \sqrt{\frac{k}{g}} \right)$$

$$\therefore t = \frac{1}{\sqrt{gk}} \left( \tan^{-1} \left( u \sqrt{\frac{k}{g}} \right) - \tan^{-1} \left( w \sqrt{\frac{k}{g}} \right) \right)$$

**Question 15 (a) (iii)**

| Criteria                    | Marks |
|-----------------------------|-------|
| • Provides correct solution | 1     |

**Sample answer:**

The second particle is at rest when  $w = 0$

$$\therefore t = \frac{1}{\sqrt{gk}} \tan^{-1} \left( u \sqrt{\frac{k}{g}} \right)$$

At this time the velocity of the first particle is

$$\frac{1}{V} = \frac{1}{u} + k \left( \frac{1}{\sqrt{gk}} \tan^{-1} \left( u \sqrt{\frac{k}{g}} \right) \right) \text{ from (i)}$$

$$\therefore \frac{1}{V} = \frac{1}{u} + \sqrt{\frac{k}{g}} \tan^{-1} \left( u \sqrt{\frac{k}{g}} \right)$$

**Question 15 (a) (iv)**

| Criteria                    | Marks |
|-----------------------------|-------|
| • Provides correct solution | 1     |

**Sample answer:**

$$\text{From (iii), } \frac{1}{V} = \frac{1}{u} + \sqrt{\frac{k}{g}} \tan^{-1} \left( u \sqrt{\frac{k}{g}} \right)$$

$$\text{for } u \text{ range, } \frac{1}{u} \approx 0, \tan^{-1} \left( u \sqrt{\frac{k}{g}} \right) \approx \frac{\pi}{2}$$

$$\therefore \frac{1}{V} \approx \sqrt{\frac{k}{g}} \cdot \frac{\pi}{2} \Rightarrow V \approx \frac{2}{\pi} \sqrt{\frac{g}{k}}$$

**Question 15 (b) (i)**

| Criteria                                   | Marks |
|--------------------------------------------|-------|
| • Provides correct solution                | 2     |
| • Obtains one of the required inequalities | 1     |

***Sample answer:***

Since for  $x \geq 0$

$$1 - x^2 \leq 1 \quad \text{ie } (1-x)(1+x) \leq 1$$

since  $(1+x) > 0$  we have  $1-x \leq \frac{1}{1+x}$

and since  $x \geq 0$  then  $1+x \geq 1$

$$\therefore \frac{1}{1+x} \leq 1$$

$$\therefore 1-x \leq \frac{1}{1+x} \leq 1$$

**Question 15 (b) (ii)**

| Criteria                                                         | Marks |
|------------------------------------------------------------------|-------|
| • Provides correct solution                                      | 2     |
| • Integrates the inequalities from part (i), or equivalent merit | 1     |

*Sample answer:*

$$\begin{aligned}\text{Since } \int_a^b \frac{1}{1+x} dx &= \left[ \ln(1+x) \right]_a^b \\ &= \ln(1+b) - \ln(1+a)\end{aligned}$$

We want  $\ln\left(1 + \frac{1}{n}\right)$  so let  $a = 0$  and  $b = \frac{1}{n}$

$$\therefore \int_0^{\frac{1}{n}} 1-x \, dx \leq \int_0^{\frac{1}{n}} \frac{1}{1+x} \, dx \leq \int_0^{\frac{1}{n}} 1 \, dx$$

$$\therefore \left[ x - \frac{x^2}{2} \right]_0^{\frac{1}{n}} \leq \left[ \ln(1+x) \right]_0^{\frac{1}{n}} \leq \left[ x \right]_0^{\frac{1}{n}}$$

$$\therefore \frac{1}{n} - \frac{1}{2n^2} \leq \ln\left(1 + \frac{1}{n}\right) \leq \frac{1}{n}$$

multiplying through by  $n$ :

$$1 - \frac{1}{2n} \leq n \ln\left(1 + \frac{1}{n}\right) \leq 1$$

**Question 15 (b) (iii)**

| Criteria                       | Marks |
|--------------------------------|-------|
| • Provides correct explanation | 1     |

**Sample answer:**

$$\text{From (ii), } 1 - \frac{1}{2n} \leq n \ln \left( 1 + \frac{1}{n} \right) \leq 1$$

$$\Rightarrow 1 - \frac{1}{2n} \leq \ln \left( 1 + \frac{1}{n} \right)^n \leq 1$$

$$\lim_{n \rightarrow \infty} \left( 1 - \frac{1}{2n} \right) \leq \lim_{n \rightarrow \infty} \ln \left( 1 + \frac{1}{n} \right)^n \leq \lim_{n \rightarrow \infty} 1$$

$$\text{ie } 1 \leq \lim_{n \rightarrow \infty} \ln \left( 1 + \frac{1}{n} \right)^n \leq 1$$

$$\Rightarrow \lim_{n \rightarrow \infty} \ln \left( 1 + \frac{1}{n} \right)^n = 1$$

$$\Rightarrow \lim_{n \rightarrow \infty} \left( 1 + \frac{1}{n} \right)^n = e$$

**Question 15 (c) (i)**

| Criteria                 | Marks |
|--------------------------|-------|
| • Provides correct proof | 1     |

**Sample answer:**

Replacing  $x$  with  $x^2$ ,  $y$  with  $y^2$  in the given inequality we have

$$\sqrt{x^2 y^2} \leq \frac{x^2 + y^2}{2}$$

$$\text{ie } xy \leq \frac{x^2 + y^2}{2}$$

$$\text{so } \sqrt{xy} \leq \sqrt{\frac{x^2 + y^2}{2}}, \text{ since } x, y > 0$$

**Question 15 (c) (ii)**

| Criteria                                                                                         | Marks |
|--------------------------------------------------------------------------------------------------|-------|
| • Provides correct proof                                                                         | 2     |
| • Uses (i) to obtain a valid expression in $a^2$ , $b^2$ , $c^2$ and $d^2$ , or equivalent merit | 1     |

**Sample answer:**

$$\begin{aligned}
 \sqrt{(ab)(cd)} &= \sqrt{ab}\sqrt{cd} \\
 &\leq \sqrt{\frac{a^2+b^2}{2}}\sqrt{\frac{c^2+d^2}{2}} \quad \text{by (i)} \\
 &\leq \sqrt{\left(\frac{a^2+b^2}{2}\right)\left(\frac{c^2+d^2}{2}\right)} \\
 &\leq \frac{1}{2}\left(\frac{a^2+b^2+c^2+d^2}{2}\right) \quad \text{by given inequality} \\
 &= \frac{a^2+b^2+c^2+d^2}{4}
 \end{aligned}$$

Taking positive square roots,

$$\sqrt[4]{abcd} \leq \sqrt{\frac{a^2+b^2+c^2+d^2}{4}}$$

**Question 16 (a) (i)**

| Criteria                                                                      | Marks |
|-------------------------------------------------------------------------------|-------|
| • Provides correct solution                                                   | 2     |
| • Shows total number of arrangements is $\binom{15}{5}$ , or equivalent merit | 1     |

**Sample answer:**

There are  $\binom{15}{5}$  ways to place 5 black counters in the grid.

There are three choices to place a black counter in each column, so there are  $3^5$  ways to place the black counters with one in each column.

$$\therefore \text{Probability} = \frac{3^5}{\binom{15}{5}} = \frac{81}{1001}$$

**Question 16 (a) (ii)**

| Criteria                                                                      | Marks |
|-------------------------------------------------------------------------------|-------|
| • Provides correct solution                                                   | 2     |
| • Shows total number of arrangements is $\binom{nq}{q}$ , or equivalent merit | 1     |

**Sample answer:**

There are  $nq$  places and we choose  $q$  places for black counters, giving  $\binom{nq}{q}$  arrangements.

There are  $n$  places for each black counter in each of  $q$  columns.

So there are  $n^q$  arrangements with one black counter in each column.

$$P_n = \frac{n^q}{\binom{nq}{q}}$$

**Question 16 (a) (iii)**

| Criteria                                                                             | Marks |
|--------------------------------------------------------------------------------------|-------|
| • Provides correct solution                                                          | 2     |
| • Obtains an expression for $P_n$ with $q$ terms in denominator, or equivalent merit | 1     |

**Sample answer:**

$$\begin{aligned}
 P_n &= \frac{n^q}{\binom{nq}{q}} \\
 &= \frac{n^q}{nq(nq-1) \dots (nq-q+1)/q!} \\
 &= \frac{q!n^q}{nq(nq-1) \dots (nq-q+1)} \\
 &= \frac{q!}{q\left(q-\frac{1}{n}\right)\left(q-\frac{2}{n}\right) \dots \left(q-\frac{q}{n}+\frac{1}{n}\right)}
 \end{aligned}$$

As  $n \rightarrow \infty$ ,

$$P_n \rightarrow \frac{q!}{q^q}$$

**Question 16 (b) (i)**

| Criteria                                           | Marks |
|----------------------------------------------------|-------|
| • Provides correct solution                        | 2     |
| • Applies De Moivre's theorem, or equivalent merit | 1     |

**Sample answer:**

By De Moivre's theorem

$$(\cos \alpha + i \sin \alpha)^{2n} = \cos(2n\alpha) + i \sin(2n\alpha)$$

By the binomial theorem,

$$\begin{aligned} (\cos \alpha + i \sin \alpha)^{2n} &= \cos^{2n} \alpha + \binom{2n}{1} \cos^{2n-1} \alpha \cdot i \sin \alpha + \binom{2n}{2} \cos^{2n-2} \alpha (i \sin \alpha)^2 + \dots \\ &\quad + \binom{2n}{2n-2} \cos^2 \alpha (i \sin \alpha)^{2n-2} + \binom{2n}{2n-1} \cos \alpha (i \sin \alpha)^{2n-1} + \binom{2n}{2n} (i \sin \alpha)^{2n} \end{aligned}$$

Equating real parts,

$$\cos(2n\alpha) = \cos^{2n} \alpha - \binom{2n}{2} \cos^{2n-2} \alpha \sin^2 \alpha + \dots + \binom{2n}{2n-2} \cos^2 \alpha \sin^{2n-2} \alpha (-1)^{n-1} + (-1)^n \sin^{2n} \alpha$$

**Question 16 (b) (ii)**

| Criteria                                                                                 | Marks |
|------------------------------------------------------------------------------------------|-------|
| • Provides correct solution                                                              | 2     |
| • Explains the connection between $T_{2n}(x)$ and $\cos(2n\alpha)$ , or equivalent merit | 1     |

**Sample answer:**

$$T_{2n}(x) = \cos(2n \cos^{-1} x)$$

put  $\alpha = \cos^{-1} x$ , so  $x = \cos \alpha$  in (i)noting that  $\sin^{2k} \alpha = (\sin^2 \alpha)^k$ 

$$= (1 - \cos^2 \alpha)^k$$

Then

$$T_{2n}(x) = x^{2n} - \binom{2n}{2} x^{2n-2} (1-x^2) + \dots + (-1)^n (1-x^2)^n$$

**Question 16 (b) (iii)**

| Criteria                                                                                                                                                                          | Marks |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|
| • Provides correct solution                                                                                                                                                       | 3     |
| • Shows that the roots of $T_{2n}(x)$ are $\cos\left(\frac{\pi}{4n}\right), \cos\left(\frac{3\pi}{4n}\right), \dots, \cos\left(\frac{(4n-1)\pi}{4n}\right)$ , or equivalent merit | 2     |
| • Obtains (or verifies) that $\cos\left(\frac{(2k-1)\pi}{4n}\right)$ are all roots of $T_{2n}(x)$ , or equivalent merit                                                           | 1     |

**Sample answer:**

$$0 = T_{2n}(x) = \cos(2n \cos^{-1} x)$$

$$\text{This occurs when } 2n \cos^{-1} x = \frac{(2k-1)\pi}{2}$$

$$\text{ie } \cos^{-1} x = \frac{(2k-1)\pi}{4n}$$

$$x = \cos\left(\frac{(2k-1)\pi}{4n}\right)$$

Consider the  $2n$  solutions with  $k = 1, 2, \dots, 2n$  ie

$$\cos\left(\frac{\pi}{4n}\right), \cos\left(\frac{3\pi}{4n}\right), \dots, \cos\left(\frac{(4n-1)\pi}{4n}\right)$$

As the angles are in the range  $0$  to  $\pi$  and are distinct these solutions are distinct. (Given that  $\cos x$  is decreasing for  $0 \leq x \leq \pi$ .)

We have  $2n$  distinct solutions of a polynomial of degree  $2n$ , and so these are all the roots.

As we have a polynomial of even degree, the product of the roots is the constant term divided by the leading term. We may find the constant term by setting  $x = 0$ , obtaining  $(-1)^n$ .

The coefficients of  $x^{2n}$  are

$$1, \quad \binom{2n}{2}, \quad \binom{2n}{4}, \quad \dots, \quad 1$$

The leading term is the sum of these and hence

$$\cos\left(\frac{\pi}{4n}\right)\cos\left(\frac{3\pi}{4n}\right)\cos\left(\frac{(4n-1)\pi}{4n}\right) = \frac{(-1)^n}{1 + \binom{2n}{2} + \binom{2n}{4} + \dots + 1}$$

In fact the denominator on the right may be simplified

$$\cos\left(\frac{\pi}{4n}\right)\cos\left(\frac{3\pi}{4n}\right)\cos\left(\frac{(4n-1)\pi}{4n}\right) = \frac{(-1)^n}{2^{2n-1}}$$

**Question 16 (b) (iv)**

| Criteria                                                                  | Marks |
|---------------------------------------------------------------------------|-------|
| • Provides correct proof                                                  | 2     |
| • Evaluates $T_{2n}\left(\frac{1}{\sqrt{2}}\right)$ , or equivalent merit | 1     |

**Sample answer:**

Put  $x = \frac{1}{\sqrt{2}}$  then  $1 - x^2 = \frac{1}{2}$

$$T_{2n}\left(\frac{1}{\sqrt{2}}\right) = \frac{1}{2^n} - \binom{2n}{2} \cdot \frac{1}{2^n} + \binom{2n}{4} \cdot \frac{1}{2^n} - \dots + (-1)^n \binom{2n}{2n} \cdot \frac{1}{2^n}$$

$$= \cos\left(2n \cos^{-1} \frac{1}{\sqrt{2}}\right) = \cos\left(\frac{n\pi}{2}\right)$$

$$\therefore 1 - \binom{2n}{2} + \binom{2n}{4} - \binom{2n}{6} + \dots + (-1)^n \binom{2n}{2n} = 2^n \cos\left(\frac{n\pi}{2}\right)$$

# 2015 HSC Mathematics Extension 2

## Mapping Grid

### Section I

| Question | Marks | Content | Syllabus outcomes |
|----------|-------|---------|-------------------|
| 1        | 1     | 3.2     | E3                |
| 2        | 1     | 2.1     | E3                |
| 3        | 1     | 1.4     | E6                |
| 4        | 1     | 7.2     | E4                |
| 5        | 1     | 2.4     | E3                |
| 6        | 1     | 4.1     | E8                |
| 7        | 1     | 8       | HE3, E9           |
| 8        | 1     | 1.7     | E6                |
| 9        | 1     | 2.2     | E3                |
| 10       | 1     | 8       | E2                |

### Section II

| Question     | Marks | Content  | Syllabus outcomes |
|--------------|-------|----------|-------------------|
| 11 (a)       | 2     | 2.1      | E3                |
| 11 (b) (i)   | 1     | 2.2      | E3                |
| 11 (b) (ii)  | 1     | 2.2      | E3                |
| 11 (b) (iii) | 1     | 2.2      | E3                |
| 11 (c)       | 2     | 7.6      | E4                |
| 11 (d)       | 2     | 3.1      | E3                |
| 11 (e)       | 3     | 1.8      | E6                |
| 11 (f) (i)   | 2     | 4.1      | PE6, E2           |
| 11 (f) (ii)  | 1     | 4.1      | E8                |
| 12 (a) (i)   | 1     | 2.2      | E3                |
| 12 (a) (ii)  | 1     | 2.2      | E3                |
| 12 (a) (iii) | 1     | 2.3      | E3                |
| 12 (b) (i)   | 3     | 7.4, 7.5 | E3, E4            |
| 12 (b) (ii)  | 1     | 7.4, 7.5 | E3, E4            |
| 12 (c) (i)   | 2     | 1.8      | E6                |
| 12 (c) (ii)  | 2     | 1.8      | E6                |
| 12 (d)       | 4     | 5.1      | E7                |
| 13 (a) (i)   | 1     | 3.2      | E3                |
| 13 (a) (ii)  | 2     | 3.2      | E3                |
| 13 (a) (iii) | 3     | 3.2      | E3                |
| 13 (b) (i)   | 1     | 8        | E2                |

| Question     | Marks | Content       | Syllabus outcomes |
|--------------|-------|---------------|-------------------|
| 13 (b) (ii)  | 2     | 5.1           | E7                |
| 13 (c) (i)   | 2     | 14.1E, 8      | HE3, E8           |
| 13 (c) (ii)  | 2     | 14.1E, 8      | HE3, E8           |
| 13 (c) (iii) | 2     | 14.1E, 8, 4.1 | HE3, E8           |
| 14 (a) (i)   | 2     | 4.1           | E8                |
| 14 (a) (ii)  | 2     | 4.1           | E8                |
| 14 (a) (iii) | 1     | 4.1           | E8                |
| 14 (b) (i)   | 1     | 7.5           | E4                |
| 14 (b) (ii)  | 2     | 7.5           | E4                |
| 14 (b) (iii) | 2     | 7.5           | E4                |
| 14 (c) (i)   | 3     | 6.3.4         | E5                |
| 14 (c) (ii)  | 2     | 6.3.4         | E5                |
| 15 (a) (i)   | 2     | 6.2.1         | E5                |
| 15 (a) (ii)  | 3     | 6.2.2         | E5                |
| 15 (a) (iii) | 1     | 6.2.1, 6.2.2  | E5                |
| 15 (a) (iv)  | 1     | 6.2.1         | E5                |
| 15 (b) (i)   | 2     | 8, 3, 1.4E    | E2, HE7           |
| 15 (b) (ii)  | 2     | 8, 1.4E, 12.5 | PE3, HE7, E2      |
| 15 (b) (iii) | 1     | 8, 1.4E       | PE3, E2           |
| 15 (c) (i)   | 1     | 8.3           | E2                |
| 15 (c) (ii)  | 2     | 8.3           | E2                |
| 16 (a) (i)   | 2     | 8             | E2                |
| 16 (a) (ii)  | 2     | 8             | E2                |
| 16 (a) (iii) | 2     | 8             | E2                |
| 16 (b) (i)   | 2     | 2.4           | E3                |
| 16 (b) (ii)  | 2     | 8             | E4                |
| 16 (b) (iii) | 3     | 7.5           | E4                |
| 16 (b) (iv)  | 2     | 7.5           | E9                |